A CONTINUAL LEARNING – BASED INTRUSION DETECTION FRAMEWORK FOR NOVEL ATTACK DETECTION IN INTERNET OF VEHICLES

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Abstract. The rapid evolution of the Internet of Vehicles (IoV) has enabled intelligent transportation systems through seamless communication among vehicles, infrastructure, and cloud platforms. However, the dynamic and distributed nature of IoV networks exposes them to advanced and continuously evolving cyber threats, making traditional intrusion detection systems inadequate due to their reliance on static models and inability to adapt to novel attacks. This paper proposes a continual learning–based intrusion detection framework designed to address these limitations by enabling adaptive and scalable security in IoV environments. The framework employs a multi-stage architecture integrating anomaly detection, attack classification, and incremental learning across roadside units and cloud infrastructure. Initially, network traffic is preprocessed to extract relevant flow-based features, and a Variational Autoencoder (VAE) is used to model normal behavior and detect anomalies without requiring labeled attack data. Identified malicious traffic is further classified using a Temporal Convolutional Neural Network (Temporal CNN) for multi-class attack detection. An open-set recognition mechanism is incorporated to identify previously unseen attacks by rejecting low-confidence predictions. These novel attack instances are then utilized in a continual learning module to incrementally update the model while preserving prior knowledge. Experimental evaluation demonstrates the framework's effectiveness in achieving robust and sustained intrusion detection performance in evolving IoV scenarios.

Keywords: Internet of Vehicles (IoV) · Intrusion Detection System (IDS) · Continual Learning · Anomaly Detection · Variational Autoencoder (VAE) · Temporal Convolutional Neural Network (Temporal CNN) · Open-Set Recognition · Novel Attack Detection · Cybersecurity · Intelligent Transportation Systems

1 Introduction

The Internet of Vehicles (IoV) represents a transformative advancement in intelligent transportation systems, enabling seamless communication among vehicles, roadside infrastructure, and cloud-based services to enhance traffic efficiency, safety, and user experience. By integrating sensing, communication, and computation capabilities, IoV facilitates real-time data exchange and decision-making across highly dynamic network environments.

However, this interconnected ecosystem significantly increases the attack surface, making vehicular networks susceptible to a wide range of cyber threats such as denial-of-service attacks, spoofing, and data manipulation. Ensuring secure and reliable communication in such environments has therefore become a critical research challenge.

Conventional Intrusion Detection Systems (IDS) primarily rely on signature-based or static machine learning approaches, which are limited in their ability to detect previously unseen attacks and adapt to evolving threat patterns. These systems often require complete retraining when new attack types emerge, leading to increased computational overhead and the problem of catastrophic forgetting, where previously learned knowledge is lost. Such limitations make traditional approaches unsuitable for the continuously evolving and real-time nature of IoV networks.

To overcome these challenges, this project proposes a Continual Learning–Based Intrusion Detection Framework tailored for IoV environments. The framework combines deep learning techniques for anomaly detection and attack classification with open-set recognition and incremental learning mechanisms. By enabling the system to detect unknown attacks and continuously update its knowledge without retraining from scratch, the proposed approach aims to provide a scalable, adaptive, and robust security solution for next-generation vehicular networks.

2 Literature Review

P. K. Tiwari et al. [1] proposed a machine learning–based intrusion detection framework for IoV that emphasizes scalability and robustness using flow-level features, achieving strong performance against known attacks but lacking effectiveness for zero-day threats. Similarly, Wei et al. [2] introduced a sustainable learning-based IDS for VANETs that adapts to dynamic vehicular environments and improves long-term stability; however, it relies heavily on labeled data and does not explicitly address novel attack detection.

Thalissas et al. [3] provided a comprehensive survey of IoV security challenges, categorizing threats across multiple communication layers, but did not propose a practical detection mechanism. In contrast, Korba et al. [4] developed Zero-X, a blockchain-enabled federated learning framework with open-set recognition for zero-day detection, offering strong adaptability but introducing significant computational and communication overhead.

Ashraf et al. [5] proposed an LSTM autoencoder–based anomaly detection model capable of identifying deviations without labeled data, though it lacks attack classification capabilities. Jain et al. [6] similarly focused on unsupervised learning for zero-day attack detection, demonstrating effectiveness in identifying unknown threats but failing to provide detailed classification for mitigation.

Guo [7] reviewed machine learning approaches for zero-day detection, highlighting challenges such as concept drift and the need for hybrid architectures, but did not present a unified solution. Mushtaq et al. [8] proposed a two-stage IDS combining autoencoders and LSTM networks for improved anomaly detection and temporal analysis, though it does not handle open-set classification or application-layer attacks.

Korium et al. [9] developed a flow-based IDS for IoV capable of detecting network-level attacks effectively; however, its performance is limited when handling subtle or application-layer threats, reinforcing the need for more comprehensive multi-stage frameworks.

3 Methodology

This project is implemented as four modules, namely: Data Preprocessing, Attack Detection Module, Attack Classification and Novel Attack Detection Module, and the Continual Learning Module.

3.1 System Architecture

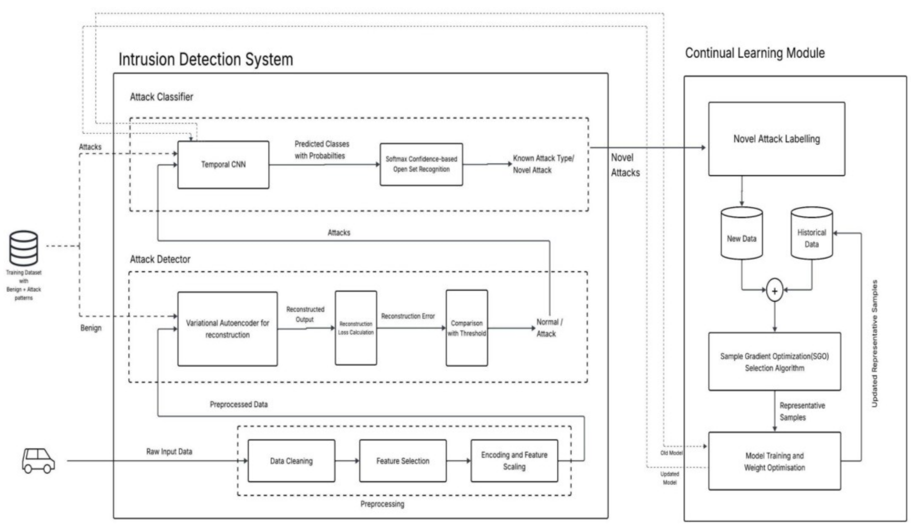

Fig. 1. Overall Architecture Diagram

The proposed system architecture presents an adaptive intrusion detection framework designed for securing Internet of Vehicles (IoV) communication environments. The architecture focuses on analyzing vehicular network traffic at the Roadside Unit (RSU) level to detect malicious activities, classify known attacks, and identify previously unseen attack patterns.

The overall system follows a multi-stage processing pipeline in which network traffic flows sequentially through preprocessing, anomaly detection, attack classification, and continual learning stages, as illustrated in Fig. 1. Initially, raw vehicular network traffic collected from IoV communication sources is provided as input to the system.

3.2 Module 1: Data Preprocessing

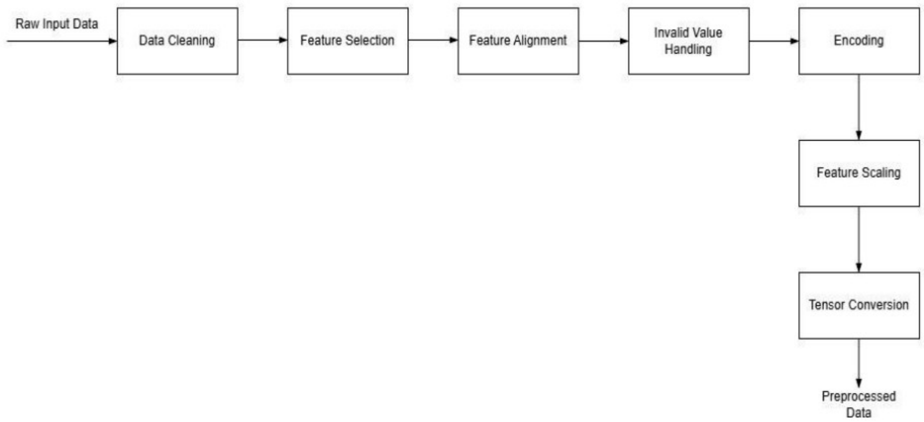

Fig. 2. Module 1: Data Preprocessing

The system begins by converting raw network traffic into a clean and normalized format suitable for deep learning. Duplicate, incomplete, and noisy records are removed, while irrelevant attributes such as IP addresses and timestamps are eliminated. Remaining features are aligned for consistency, and invalid values like NaN and outliers are handled. Categorical data is encoded numerically, and all features are scaled to a common range. Finally, the processed data is transformed into tensor format for efficient model input.

3.3 Module 2: Attack Detection

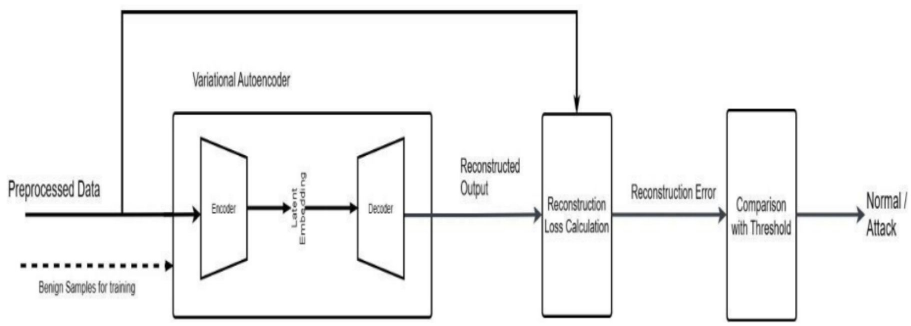

Fig. 3. Module 2: Attack Detection

The attack detection module uses a Variational Autoencoder (VAE) to identify malicious network traffic by learning normal behavior patterns. Trained on benign data, it encodes inputs into a latent space and reconstructs them. During inference, reconstruction error is calculated for each sample, and those exceeding a predefined threshold are flagged as anomalies, indicating potential intrusion attempts.

Algorithm 1 VAE-based Attack Detection

Input: Preprocessed traffic features
1: Train VAE using normal traffic
2: **for** each incoming network flow **do**
3:     Compute reconstruction error
4:     **if** error > threshold **then**
5:         Label flow as **Attack**
6:     **else**
7:         Label flow as **Normal**
8:     **end if**
9: **end for**

3.4 Module 3: Attack Classification and Novel Attack Detection

The attack classification and novel attack detection module analyzes malicious samples identified by the VAE to determine attack types and detect unseen threats. Samples are processed using a Temporal CNN, which captures sequential patterns and outputs class probabilities.

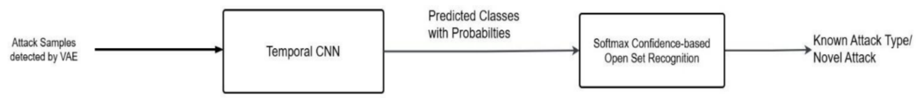

Fig. 4. Module 3: Attack Classification and Novel Attack Detection

These probabilities are evaluated using a softmax-based open-set recognition mechanism, where high-confidence predictions are assigned to known classes, and low-confidence ones are flagged as novel attacks. This approach enables accurate classification while supporting detection of emerging threats.

Algorithm 2 Temporal CNN-based Attack Classification and Novel Attack Detection

Input: Attack samples detected by VAE

```
Load trained Temporal CNN classifier
for each attack sample do
    Extract temporal feature sequence
    Compute class probabilities using Softmax
    if maximum probability < confidence threshold then
        Label as Novel Attack
    else
        Label as Known Attack Class
    end if
end for
```

3.5 Module 4: Continual Learning Module

The continual learning module enables the IDS to adapt to evolving attacks while preserving prior knowledge. Novel attacks identified through open-set recognition are validated and labeled, then combined with selected historical data for balanced learning. A Sample Gradient Optimization (SGO)–based method selects the most informative samples to prevent catastrophic forgetting and reduce retraining cost. These samples are used for incremental model updates, initialized from the previous model, and the updated model is fed back to improve future detection and classification.

Algorithm 3 SGO-based Continual Learning for Novel Attack Adaptation

Input: Labeled novel attack samples, historical attack data
Output: Updated intrusion detection model

```
Initialize model with previous trained weights
Combine novel attack data with historical data
Apply Sample Gradient Optimization (SGO)
Train model using selected samples
Update model weights
Store selected samples in memory buffer
Output updated model
```

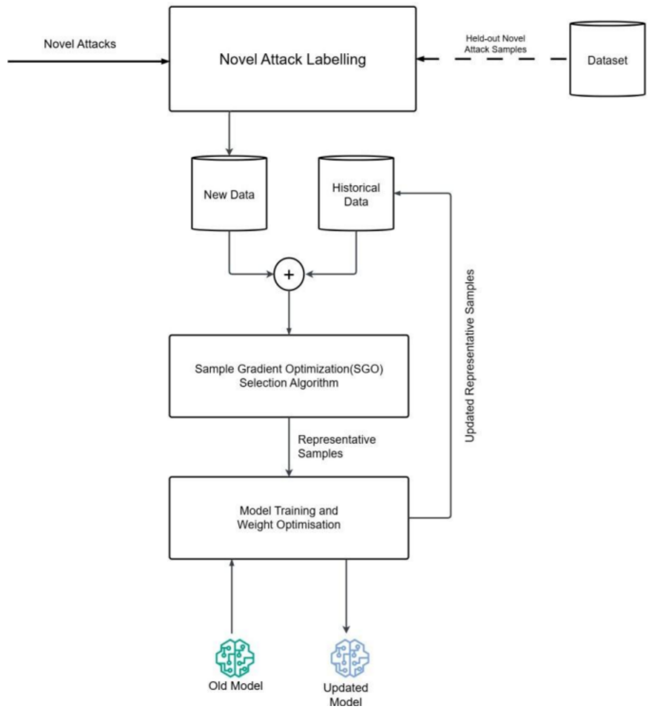

Fig. 5. Module 4: Continual Learning Module

Algorithm 4 Sample Gradient Optimization (SGO)

Input: Batch data for current task, neural network model, hyperparameters
Output: Selected set of representative training samples

1: **for** each batch in current task **do**
2: Evaluate sample importance based on gradient contribution
3: Compute gradient information for samples in the batch
4: Compare gradient contribution across batches
5: Select samples with higher contribution
6: **end for**
7: Store selected samples in memory buffer
8: Return selected representative samples

4 Performance Metrics

Table 1. Attack Detection Performance

Metric	**Accuracy**	**Precision**	**Recall**	**F1-Score**
Value	94%	97%	95%	94%

Table 2. Attack Classification Performance

Metric	Acc	Prec	Rec	F1	NADR	FRR
Value	95%	97%	95%	95%	100%	0.26%

Table 3. Continual Learning Performance

Metric	Acc (%)	Prec (%)	Rec (%)	F1 (%)
Value	87.81	84.37	87.81	85.64

5 Conclusion and Future Work

This project presents an intelligent intrusion detection framework for IoV using machine learning and continual learning to address evolving cyber threats. The system effectively detects multiple attack types while adapting to new patterns without retraining. Its scalability, stability, and strong performance make it suitable for securing next-generation intelligent transportation systems in real-time environments.

Future work can focus on real-time deployment with reduced latency and lightweight models for edge devices. Enhancements such as automated labeling, adaptive thresholds, and explainable AI can improve zero-day detection and transparency. Evaluating on diverse datasets and integrating hybrid models can further strengthen generalization and overall system robustness.

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